

ARNAB BHATTACHARYYA

Gopalpur, West Bengal, India

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SUMMARY

Full-Stack Engineer with a B.Tech in AI & ML and experience building scalable web applications, REST APIs, and AI-powered solutions using TypeScript, Python, Node.js, LangChain, and RAG.

EDUCATION

Brainware University

B.Tech in CSE (Artificial Intelligence & Machine Learning) — CGPA: 9.53

Aug 2021 – July 2025

Barasat, West Bengal

EXPERIENCE

Capsitech IT Solution

July 2025 – Present

Assistant System Engineer (Software Development) – L1

Jodhpur, Rajasthan – On-site

- Engineered 10+ full-stack features using TypeScript across Node.js and React/Next.js, building REST APIs, real-time features (Socket.io), and optimizing data workflows with MongoDB for improved query performance and reduced latency.
- Developed SEO-optimized UI components for CapsiCore, improving crawlability via structured metadata, routing strategies, and form handling using Zod and Formik.
- Built and maintained CRM systems (.NET) for Acting Office, contributing to UK compliance workflows with cross-functional collaboration.

Aeka Advisors

April 2025 – June 2025

Web Developer Intern

Kolkata, West Bengal – On-site

- Built a 10+ component SvelteKit UI system, reducing duplication and improving dev speed.
- Centralized application state with Svelte stores, removing prop drilling across 5+ shared UI components.
- Enhanced UI responsiveness for 6+ screen breakpoints, ensuring smooth layouts on mobile, tablet, and desktop.

Cehpoint E-Learning & Cyber Security Solutions

Oct 2024 – Feb 2025

MERN Stack & AI Integration Intern

Remote

- Contributed to full-stack development using the MERN stack (MongoDB, Express.js, React.js, Node.js).
- Enabled secure authentication and real-time database capabilities via Firebase, validated with 100+ users.
- Assisted in embedding AI-driven enhancements to elevate automation and user experience.

PROJECTS

Nera | Next.js, Fastify, PostgreSQL, Minio

- Built a secure file management system with encrypted storage using S3-compatible object storage (Minio).
- Implemented JWT authentication (HTTP-only cookies) and role-based access control for secure file operations.
- Designed Fastify REST APIs with PostgreSQL for file upload, search, and nested folder management.

Coditor | Next.js, ASP.NET, Docker

- Developed a containerized code execution platform for secure multi-language code execution.
- Used Docker-based isolation to safely run user code and capture execution output.
- Implemented CI/CD pipelines and automated cleanup workflows for reliable deployments.

Talk-to-PDF | Python, LangChain, ChromaDB, Sentence-Transformers

- Built a RAG-based CLI application for querying PDF documents using natural language.
- Implemented document chunking, embeddings, and semantic retrieval using LangChain and ChromaDB.
- Generated source-aware responses with retrieved context and similarity-based search.

SKILLS

Languages JavaScript, TypeScript, Python, C

Domains Web Development, Object Oriented Programming (OOP), DBMS, Data Structures & Algorithms (DSA)

Technologies React.js, Next.js, SvelteKit, Tailwind CSS, Node.js, Express.js, .NET, REST APIs, JWT Authentication, MongoDB, MySQL, Zod, Formik

DevOps & Cloud Git, GitHub, GitHub Actions (CI/CD), Docker, Vercel, Render, S3-compatible storage (Minio)

AI/ML & NLP LangChain, Retrieval-Augmented Generation (RAG), Vector Databases, NLP, Scikit-learn, NLTK, SpaCy

AWARDS & CERTIFICATIONS

- 5th Place, HustleX – IIM Lucknow [Certificate]
- Top 22, Ideathon – IIM Bangalore [Certificate]